Telecom Customer Analysis Dashboard

Business scenario: A telecom company provides mobile, broadband, and internet services to customers. The company wants to understand its revenue, customer usage, active customers, and customers who have stopped using its services.

1. Project objective

To analyze telecom customer data and identify:

- Total and active customers.
- Customers who have discontinued services.
- Revenue generated by different service types.
- Which subscription plans are popular.
- Which contract types have more customer churn.
- Which regions have more customers leaving.

Project description: We'll create a telecom customer dataset in Snowflake, perform SQL analysis, connect it to Power BI, create five calculated columns using Power Query, build DAX measures, and develop a dashboard with KPI cards, charts, and slicers.

2. Snowflake setup

Create database, schema, and warehouse

```
CREATE DATABASE IF NOT EXISTS UDAY_TELECOM_DB;

CREATE SCHEMA IF NOT EXISTS UDAY_TELECOM_DB.TELECOM_SCHEMA;

CREATE WAREHOUSE IF NOT EXISTS UDAY_TELECOM_WH
WITH
    WAREHOUSE_SIZE = 'X-SMALL'
    AUTO_SUSPEND = 60
    AUTO_RESUME = TRUE
    INITIALLY_SUSPENDED = TRUE;

USE WAREHOUSE UDAY_TELECOM_WH;
USE DATABASE UDAY_TELECOM_DB;
USE SCHEMA TELECOM_SCHEMA;
```



Create the table

```sql
CREATE OR REPLACE TABLE TELECOM_CUSTOMERS (
    CUSTOMER_ID VARCHAR(10),
    CUSTOMER_NAME VARCHAR(50),
    JOIN_DATE DATE,
    CITY VARCHAR(30),
    REGION VARCHAR(20),
    SERVICE_TYPE VARCHAR(30),
    PLAN_NAME VARCHAR(30),
    CONTRACT_TYPE VARCHAR(20),
    MONTHLY_CHARGES NUMBER(10,2),
    DATA_USAGE_GB NUMBER(10,2),
    CALL_MINUTES NUMBER(10,0),
    SUPPORT_TICKETS NUMBER(5,0),
    PAYMENT_METHOD VARCHAR(20),
    CUSTOMER_STATUS VARCHAR(20)
);
```

Results (just now)

Table | Chart | Pivot — 1 row — 140ms

	status
1	Table TELECOM_CUSTOMERS successfully created.

Insert sample records

```sql
INSERT INTO TELECOM_CUSTOMERS VALUES
('T001','Ravi','2024-01-15','Hyderabad','South','Mobile','Premium','Yearly',799,45,650,1,'UPI','Active'),
('T002','Anjali','2024-02-10','Chennai','South','Broadband','Basic','Monthly',499,120,0,4,'Card','Churned'),
('T003','Kiran','2024-03-05','Pune','West','Fiber','Premium','Two-Year',1299,250,0,1,'UPI','Active'),
('T004','Suresh','2024-03-20','Delhi','North','Mobile','Basic','Monthly',399,18,420,5,'Cash','Churned'),
('T005','Priya','2024-04-12','Kolkata','East','Broadband','Standard','Yearly',699,180,0,2,'Card','Active'),
('T006','Manoj','2024-05-08','Hyderabad','South','Mobile','Standard','Monthly',599,32,520,3,'UPI','Active'),
('T007','Sneha','2024-05-22','Mumbai','West','Fiber','Premium','Two-Year',1499,300,0,0,'Card','Active'),
('T008','Arjun','2024-06-14','Bangalore','South','Broadband','Basic','Monthly',449,95,0,6,'Cash','Churned'),
('T009','Divya','2024-07-03','Jaipur','North','Mobile','Premium','Yearly',899,65,780,1,'UPI','Active'),
('T010','Rahul','2024-07-19','Chennai','South','Fiber','Standard','Monthly',999,210,0,3,'Card','Active'),
('T011','Meena','2024-08-11','Pune','West','Mobile','Basic','Monthly',349,15,300,7,'Cash','Churned'),
('T012','Vikram','2024-08-26','Delhi','North','Broadband','Premium','Two-Year',1199,240,0,1,'UPI','Active'),
('T013','Lakshmi','2024-09-09','Hyderabad','South','Fiber','Standard','Yearly',899,190,0,2,'Card','Active'),
('T014','Ajay','2024-09-21','Kolkata','East','Mobile','Standard','Monthly',549,38,590,4,'UPI','Churned'),
('T015','Geetha','2024-10-04','Mumbai','West','Broadband','Premium','Yearly',999,220,0,0,'Card','Active'),
('T016','Naveen','2024-10-18','Bangalore','South','Mobile','Basic','Monthly',399,20,350,5,'Cash','Churned'),
('T017','Harish','2024-11-07','Chennai','South','Fiber','Premium','Two-Year',1399,280,0,1,'UPI','Active'),
```

Results (just now)

Table | Chart | Pivot — 1 row — 735ms

	number of rows inserted
1	25

Verify the data

```sql
SELECT * FROM TELECOM_CUSTOMERS;
```

Results (just now)

Table | Chart | Pivot | Next steps — 25 rows — 59ms

	CUSTOMER_ID	CUSTOMER_NAME	JOIN_DATE	CITY	REGION	SERVICE_TYPE	PLAN_NAME	CONTRACT_TYPE	MONTHLY_CHARGE
	T001 4.0% T002 4.0% +23 more	Ajay 4.0% Anjali 4.0% +23 more	1/15/... 3/18/...	Chen... 16.0% Hyde... 16.0% +6 more	So... 44.0% West 24.0% +2 more	Mobile 40.0% Broadband 32.0% +1 more	Premium 44.0% Basic 28.0% +1 more	Monthly 44.0% Yearly 32.0% +1 more	349
1	T001	Ravi	2024-01-15	Hyderabad	South	Mobile	Premium	Yearly	79
2	T002	Anjali	2024-02-10	Chennai	South	Broadband	Basic	Monthly	49

```
65
66  SELECT COUNT(*) AS TOTAL_CUSTOMERS
67  FROM TELECOM_CUSTOMERS;
```

Results (just now)

Table | Chart | Pivot

	# TOTAL_CUSTOMERS
1	25

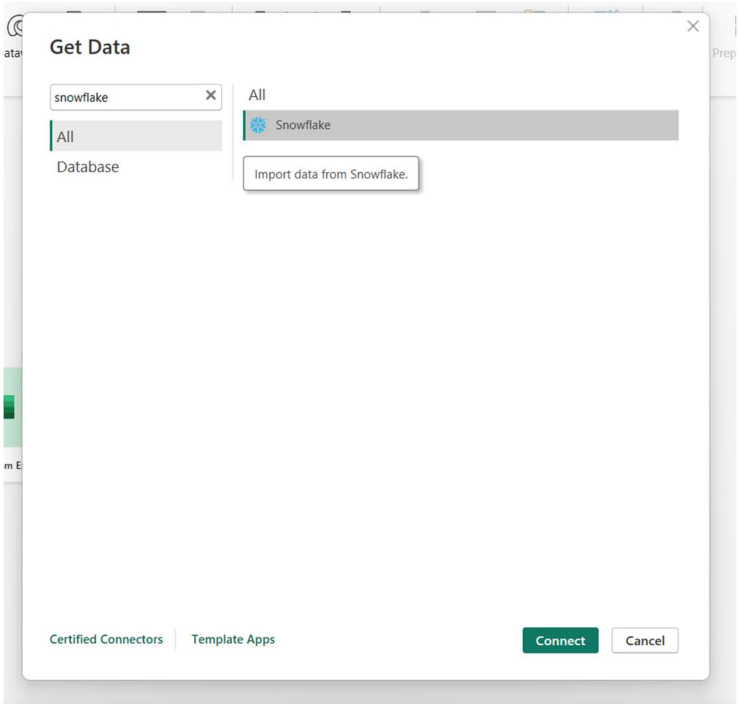
```
69
70  SELECT
71      SERVICE_TYPE,
72      COUNT(DISTINCT CUSTOMER_ID) AS TOTAL_CUSTOMERS
73  FROM TELECOM_CUSTOMERS
74  GROUP BY SERVICE_TYPE
75  ORDER BY TOTAL_CUSTOMERS DESC;
```

Results (just now)

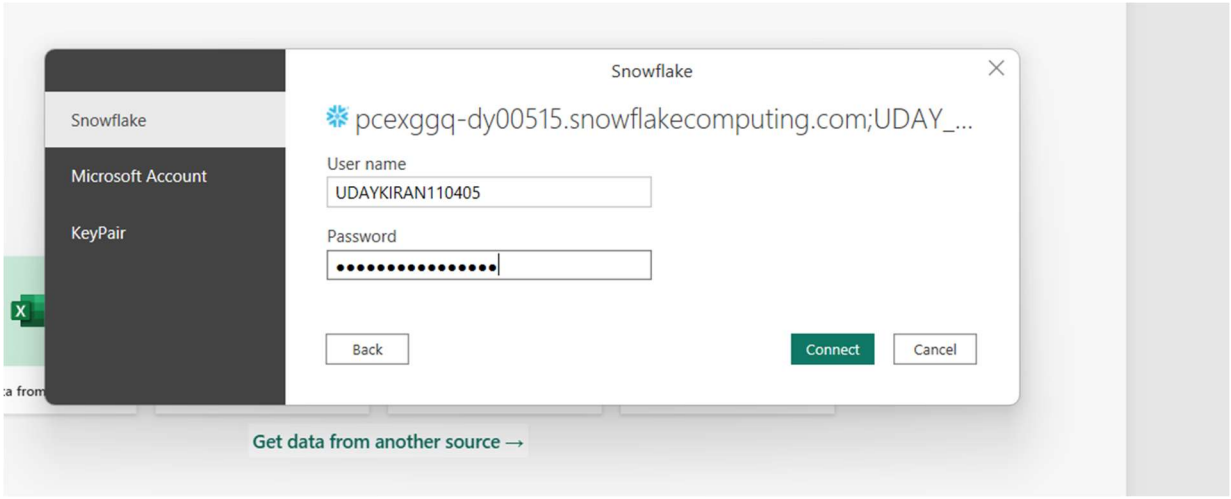
Table | Chart | Pivot — 3 rows — 527ms

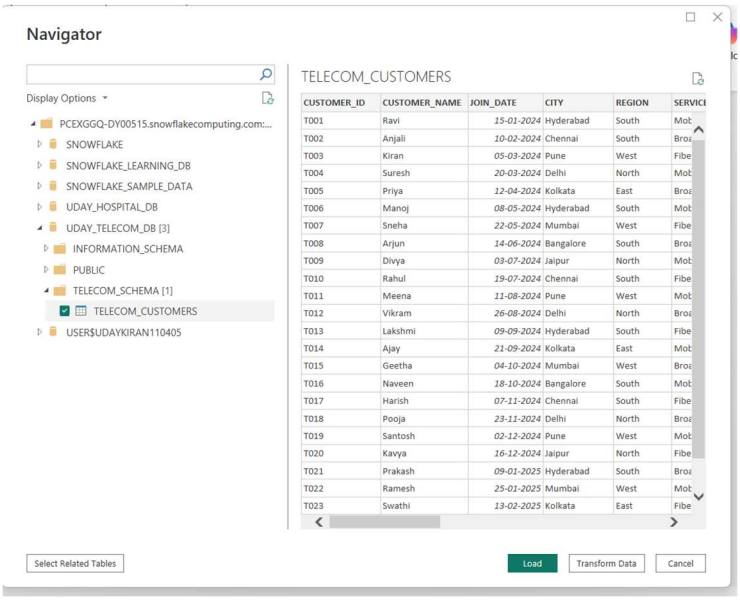
	A SERVICE_TYPE	# TOTAL_CUSTOMERS
1	Mobile	10
2	Broadband	8
3	Fiber	7

Connecting to the Power BI :

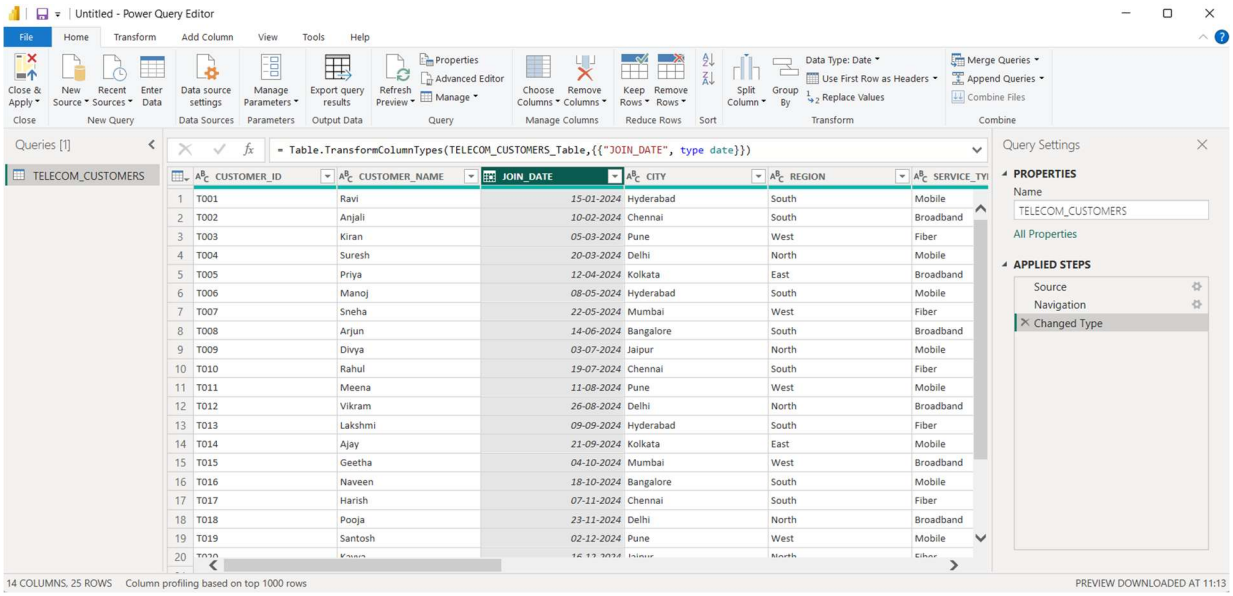



Snowflake
Server
PCEXGGQ-DY00515.snowflakecomputing.com
Warehouse
UDAY_TELECOM_WH
Advanced options
OK
Cancel

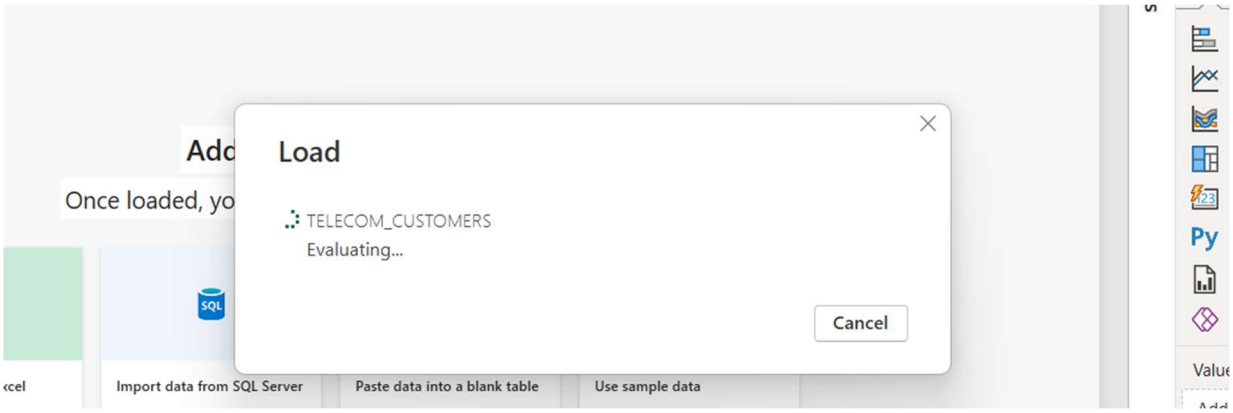

Snowflake
Snowflake
Microsoft Account
KeyPair
pcexggq-dy00515.snowflakecomputing.com;UDAY_...
User name
UDAYKIRAN110405
Password
Back
Connect
Cancel
Get data from another source →


Navigator
Display Options
PCEXGGQ-DY00515.snowflakecomputing.com:...
SNOWFLAKE
SNOWFLAKE_LEARNING_DB
SNOWFLAKE_SAMPLE_DATA
UDAY_HOSPITAL_DB
UDAY_TELECOM_DB [3]
INFORMATION_SCHEMA
PUBLIC
TELECOM_SCHEMA [1]
TELECOM_CUSTOMERS
USER$UDAYKIRAN110405
TELECOM_CUSTOMERS
CUSTOMER_ID CUSTOMER_NAME JOIN_DATE CITY REGION SERVICE
T001 Ravi 15-01-2024 Hyderabad South Mob
T002 Anjali 10-02-2024 Chennai South Broa
T003 Kiran 05-03-2024 Pune West Fibe
T004 Suresh 20-03-2024 Delhi North Mob
T005 Priya 12-04-2024 Kolkata East Broa
T006 Manoj 08-05-2024 Hyderabad South Mob
T007 Sneha 22-05-2024 Mumbai West Fibe
T008 Arjun 14-06-2024 Bangalore South Broa
T009 Divya 03-07-2024 Jaipur North Mob
T010 Rahul 19-07-2024 Chennai South Fibe
T011 Meena 11-08-2024 Pune West Mob
T012 Vikram 26-08-2024 Delhi North Broa
T013 Lakshmi 09-09-2024 Hyderabad South Fibe
T014 Ajay 21-09-2024 Kolkata East Mob
T015 Geetha 04-10-2024 Mumbai West Broa
T016 Naveen 18-10-2024 Bangalore South Mob
T017 Harish 07-11-2024 Chennai South Fibe
T018 Pooja 23-11-2024 Delhi North Broa
T019 Santosh 02-12-2024 Pune West Mob
T020 Kavya 16-12-2024 Jaipur North Fibe
T021 Prakash 09-01-2025 Hyderabad South Broa
T022 Ramesh 25-01-2025 Mumbai West Mob
T023 Swathi 13-02-2025 Kolkata East Fibe
Select Related Tables
Load
Transform Data
Cancel

Checking the Data Types :



Loading the data :

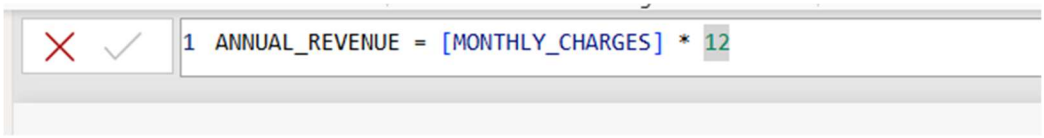


Create five calculated columns in Power Query

1.ANNUAL_REVENUE

[MONTHLY_CHARGES] * 12

Estimates the annual subscription revenue from the customer's monthly charges.



2. CUSTOMER_TENURE

CUSTOMER_TENURE =

DATEDIFF(

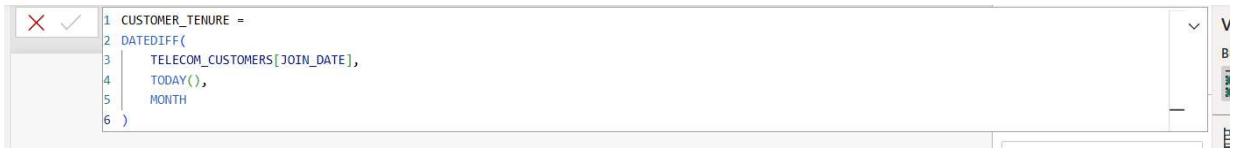
TELECOM_CUSTOMERS[JOIN_DATE],

TODAY(),

MONTH

)

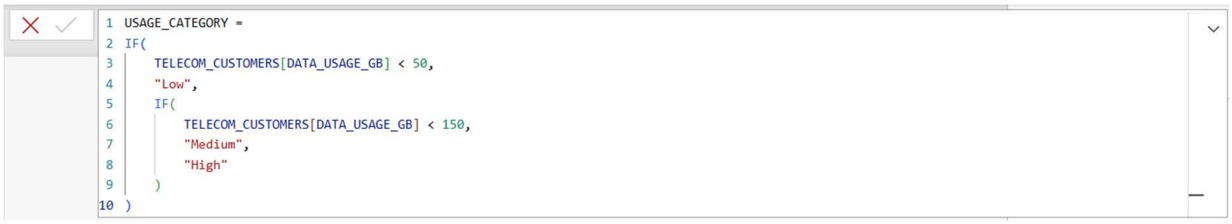
Calculates the number of months since the customer joined. It updates when the query is refreshed.



3. USAGE_CATEGORY

```
USAGE_CATEGORY =
IF(
   TELECOM_CUSTOMERS[DATA_USAGE_GB] < 50,
   "Low",
   IF(
      TELECOM_CUSTOMERS[DATA_USAGE_GB] < 150,
      "Medium",
      "High"
   )
)
```

Categorizes customers based on their monthly data usage.



```
USAGE_CATEGORY =
IF(
    TELECOM_CUSTOMERS[DATA_USAGE_GB] < 50,
    "Low",
    IF(
        TELECOM_CUSTOMERS[DATA_USAGE_GB] < 150,
        "Medium",
        "High"
    )
)
```

4.SUPPORT_CATEGORY

```
SUPPORT_CATEGORY =
IF(
   TELECOM_CUSTOMERS[SUPPORT_TICKETS] <= 2,
   "Low",
   IF(
      TELECOM_CUSTOMERS[SUPPORT_TICKETS] <= 4,
      "Medium",
      "High"
   )
)
```



```
SUPPORT_CATEGORY =
IF(
    TELECOM_CUSTOMERS[SUPPORT_TICKETS] <= 2,
    "Low",
    IF(
        TELECOM_CUSTOMERS[SUPPORT_TICKETS] <= 4,
        "Medium",
        "High"
    )
)
```

Classifies customers according to the number of support tickets raised.

5. CHURN_FLAG

```
CHURN_FLAG =
IF(
   TELECOM_CUSTOMERS[CUSTOMER_STATUS] = "Churned",
   1,
   0
)
```



```
CHURN_FLAG =
IF(
    TELECOM_CUSTOMERS[CUSTOMER_STATUS] = "Churned",
    1,
    0
)
```

Converts customer status into a numeric value: 1 for churned and 0 for active.

Checking :

ANNUAL_REVENUE	CUSTOMER_TENURE	USAGE_CATEGORY	SUPPORT_CATEGORY	CHURN_FLAG
9588	34	Low	Low	0
5988	33	Medium	Medium	1
15588	32	High	Low	0
4788	32	Low	High	1
8388	31	High	Low	0
7188	30	Low	Medium	0
17988	30	High	Low	0
5388	29	Medium	High	1
10788	28	Medium	Low	0
11988	28	High	Medium	0
4188	27	Low	High	1
14388	27	High	Low	0
10788	26	High	Low	0
6588	26	Low	Medium	1
11988	25	High	Low	0
4788	25	Low	High	1
16788	24	High	Low	0
7788	24	High	Medium	0
10188	23	Medium	Low	0
9588	23	Medium	High	1
14388	22	High	Low	0
7788	22	Low	Low	0
15588	21	High	Medium	1
5988	20	Medium	Low	0
11988	20	Medium	Low	0

Create DAX measures

1.Total Customers

Total Customers =
DISTINCTCOUNT(TELECOM_CUSTOMERS[CUSTOMER_ID])

Counts unique customers.



2. Active Customers

Active Customers =
CALCULATE(
  [Total Customers],
  TELECOM_CUSTOMERS[CUSTOMER_STATUS] = "Active"
)

Counts customers who are currently active.



3. Churned Customers

Churned Customers =

CALCULATE(

[Total Customers],

TELECOM_CUSTOMERS[CUSTOMER_STATUS] = "Churned"

)



Counts customers who have discontinued their services.

4. Churn Rate

Churn Rate =
DIVIDE(
[Churned Customers],
[Total Customers],
0
)



Calculates the percentage of customers who have churned. Format this measure as Percentage.

5. Total Monthly Revenue

Total Monthly Revenue =
SUM(TELECOM_CUSTOMERS[MONTHLY_CHARGES])



Calculates the total monthly subscription revenue represented by the dataset.

6. Average Monthly Revenue

```
Average Monthly Revenue =
DIVIDE(
   [Total Monthly Revenue],
   [Total Customers],
   0
)
```



Calculates the average monthly charge per customer.

7. Total Annual Revenue

```
Total Annual Revenue =
SUM(TELECOM_CUSTOMERS[ANNUAL_REVENUE])
```



Calculates the annualized subscription revenue based on monthly charges.

8. Revenue Lost Due to Churn

```
Revenue Lost Due to Churn =
CALCULATE(
   [Total Monthly Revenue],
   TELECOM_CUSTOMERS[CUSTOMER_STATUS] = "Churned"
)
```

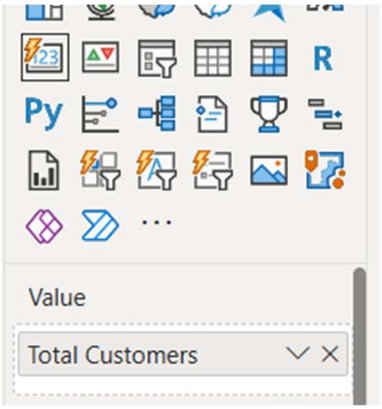


Calculates the monthly subscription revenue associated with churned customers. This represents estimated monthly revenue at risk or lost, not a verified historical loss.

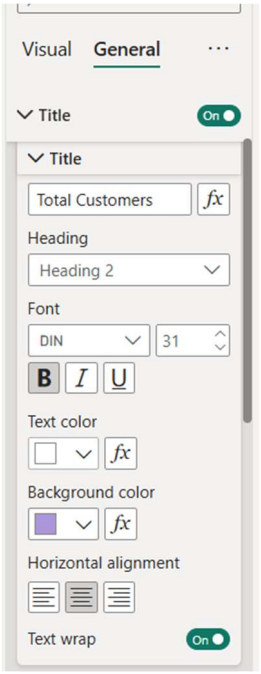
Create the visualizations

Displaying the measures as the KPI Cards :

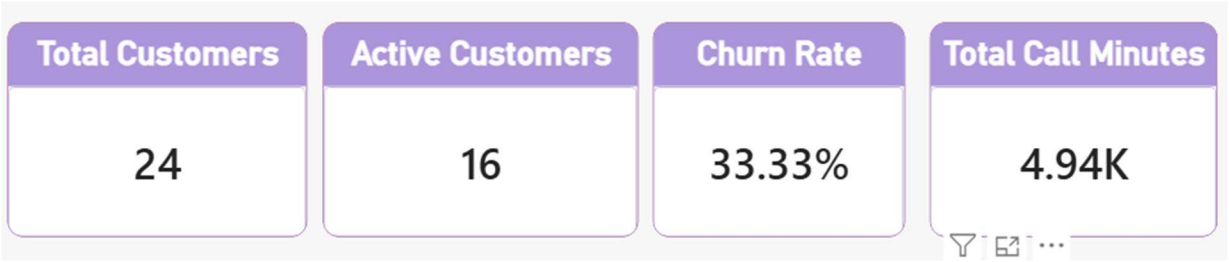




Styles:



Creating the remaining KPI Cards:



Charts:

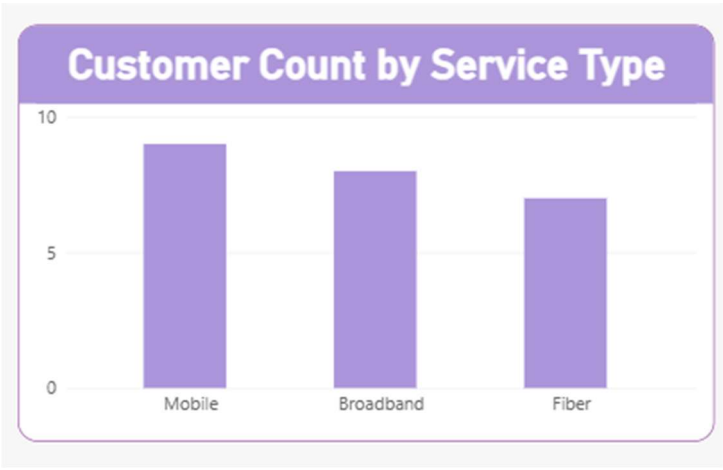
1. Customer Count by Service Type

- Visual: Clustered Column Chart
- X-axis: SERVICE_TYPE
- Y-axis: Total Customers

Business insight: Shows how customers are distributed across Mobile, Broadband, and Fiber.

Interpretation: Identifies the service type with the largest customer base.

Recommendation: Focus on retaining customers in the largest service categories and improving adoption in smaller categories.



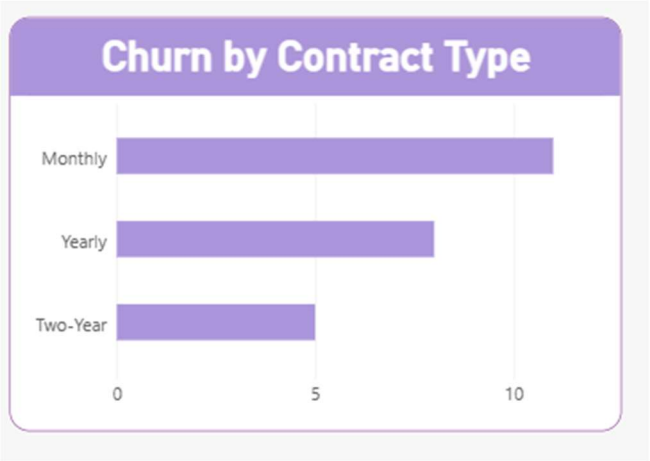
2. Churn by Contract Type

- Visual: Stacked Bar Chart
- Y-axis: CONTRACT_TYPE
- X-axis: Total Customers
- Legend: CUSTOMER_STATUS

Business insight: Compares active and churned customers across monthly, yearly, and two-year contracts.

Interpretation: Reveals which contract type has a larger share of churned customers.

Recommendation: Consider retention offers or incentives for contract types with higher churn.



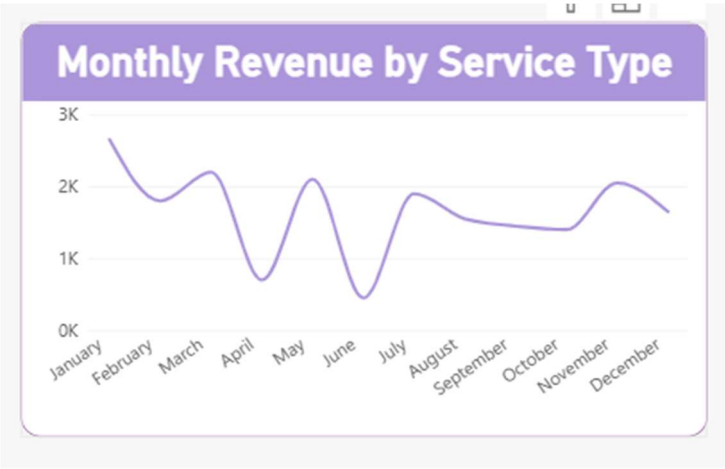
3. Monthly Revenue by Service Type

- Visual: Line Chart
- X-axis: JOIN_DATE (Month)
- Y-axis: Total Monthly Revenue
- Legend: SERVICE_TYPE

Business insight: Compares revenue from different service types over customer joining months.

Interpretation: Shows how the revenue contribution of each service type changes across the recorded joining periods.

Recommendation: Use the pattern to investigate which service types attract higher-value subscriptions.



Note: This chart shows revenue by customer joining month, not historical monthly billing revenue. The dataset does not contain monthly billing records.

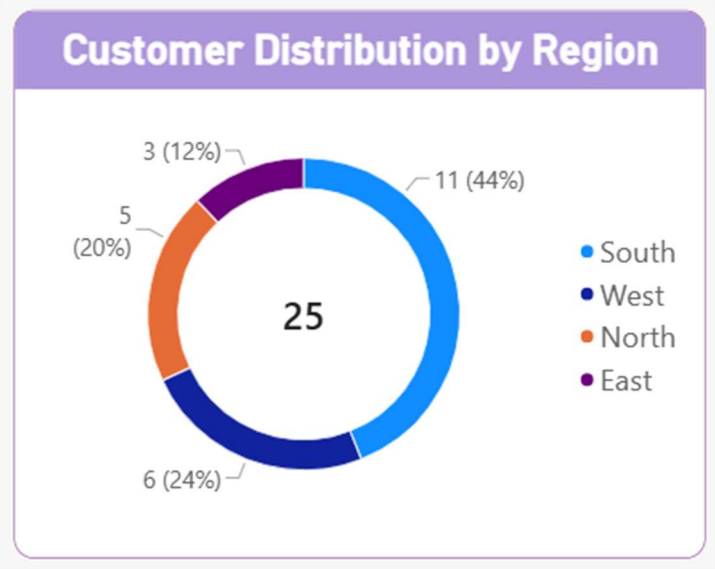
4. Customer Distribution by Region

- Visual: Donut Chart
- Legend: REGION
- Values: Total Customers

Business insight: Shows the geographical distribution of customers.

Interpretation: Identifies regions with the largest and smallest customer bases.

Recommendation: Focus marketing and service expansion on regions with growth potential.



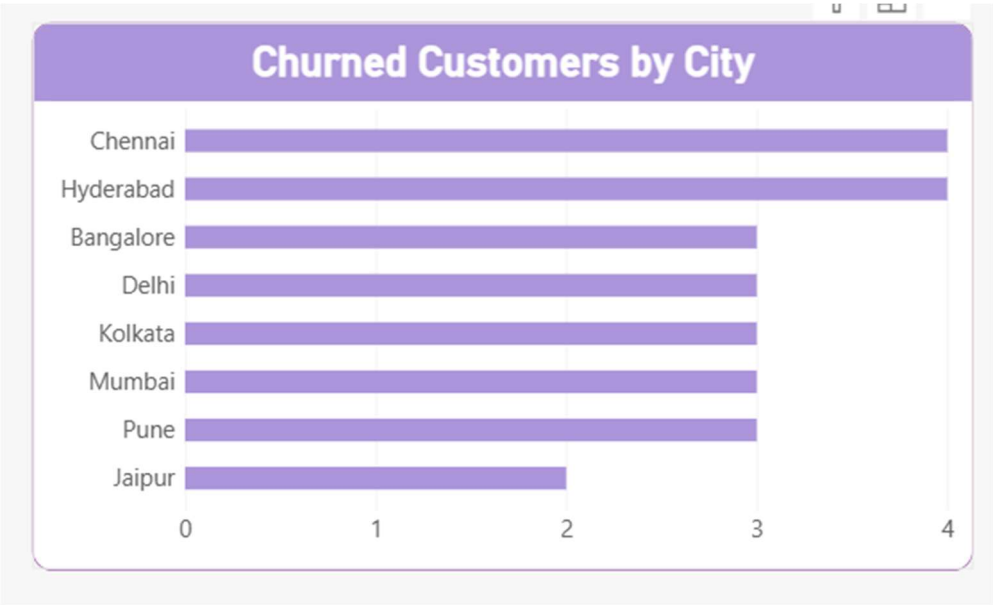
5. Churned Customers by City

- Visual: Clustered Bar Chart
- Y-axis: CITY
- X-axis: Churned Customers

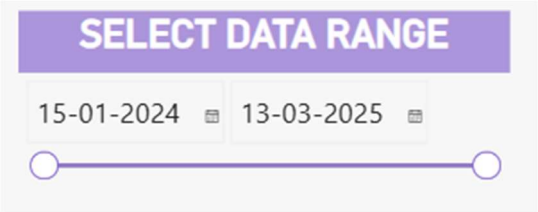
Business insight: Identifies cities with the highest number of churned customers.

Interpretation: Shows where customer retention needs more attention.

Recommendation: Investigate service quality, network coverage, and customer complaints in those cities.

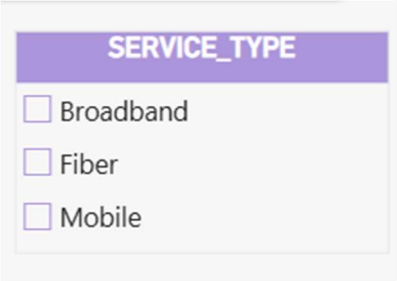


Slicer1 :

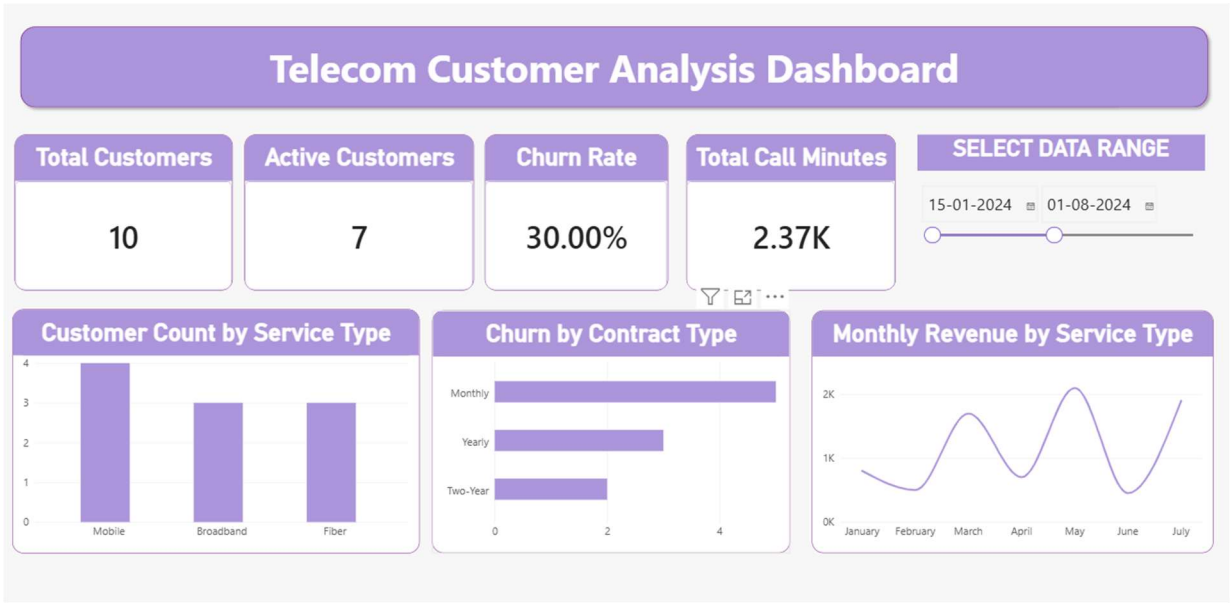


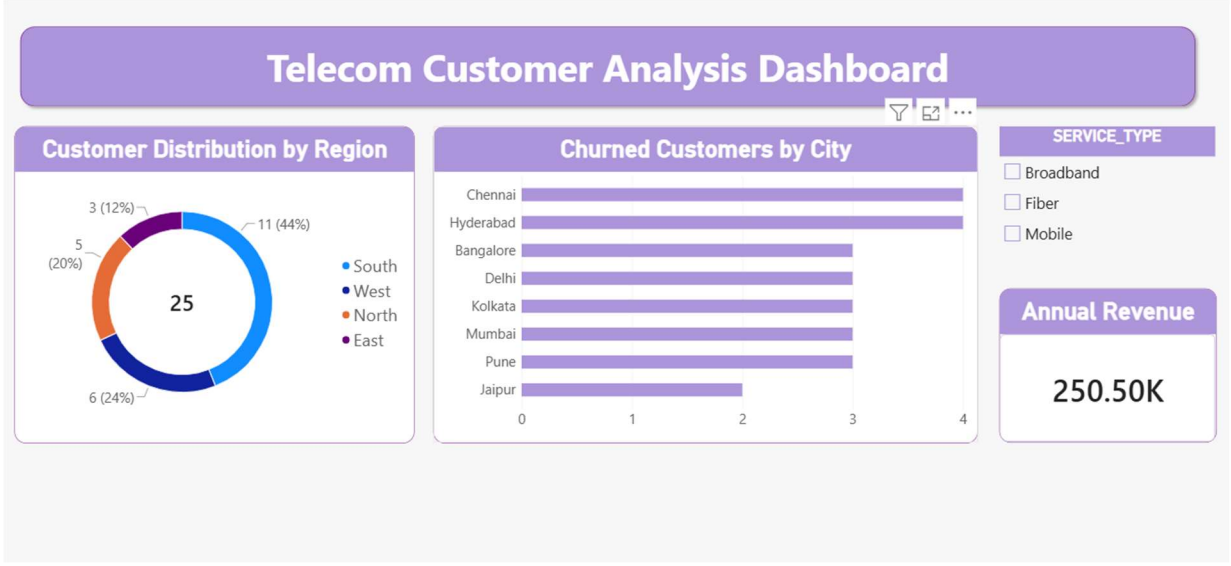
use the Between date slicer to filter customers by joining date.

Slicer 2:

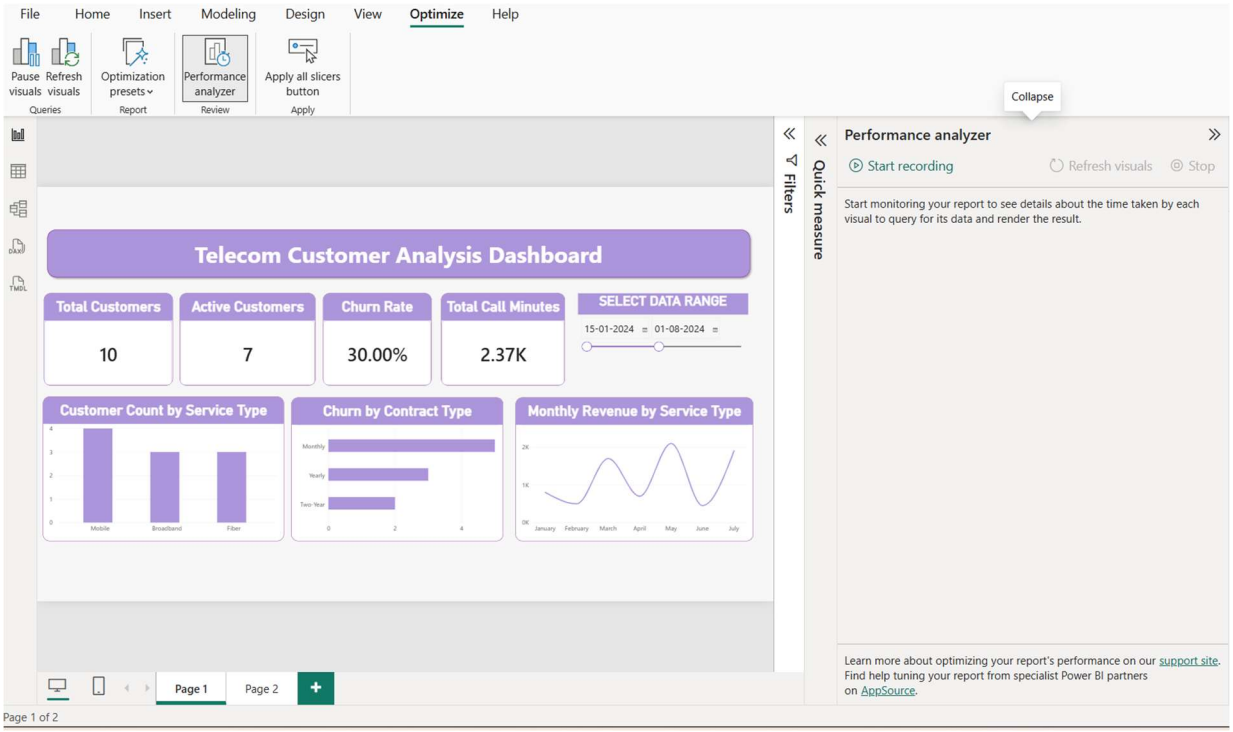


Final Dashboard :





Performance Analyzer:



Click The Start Recording and Refresh :

Results:

